

# MARY ELLEN CHAFFIN

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## SUMMARY

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Senior front-end engineer who builds interfaces people enjoy using. Not just functional --- delightful. Things are where users expect them. Nothing distracts from getting things done. Strong in React and TypeScript, with a practical eye for performance --- focused on what users experience.

## CAPABILITIES

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- React, TypeScript, JavaScript
- State management and async data flows
- UX writing and interface clarity
- Data visualization: Highcharts, D3
- UI performance and front-end architecture
- REST and GraphQL API integration
- Design systems: Material UI, Cloudscape
- Unit testing with Vitest
- Cross-functional collaboration with design and backend teams

## EXPERIENCE

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### Senior Front-End Engineer

**Amazon Web Services** (via 24Seven contract) - Remote | *Aug 2024 – Feb 2026*

- Shipped two production applications from zero, including a tool delivered to AWS Console users in under four months.
- Owned the full front-end from architecture to shipped experience on a greenfield service developed with React, TypeScript, TanStack Query, Vitest, and Cloudscape.
- Built a centralized state system driving routing, URL construction, and data fetching across a complex multi-page application eliminating stale filter bugs and keeping the URL reliably in sync with what the user sees.
- Built a data visualization layer using D3 and Highcharts, with five chart types sharing common infrastructure for responsive sizing, pan/zoom, loading states, and error handling.
- Designed a three-tier in-app guidance system (reference docs, walkthroughs, and interactive tutorials) tuned to different user skill levels and moments of need.
- Wrote UX copy throughout both applications (labels, empty states, errors, onboarding) treating language as part of the product experience, not a finishing step.
- Collaborated directly with backend engineers to shape API contracts, flag UX gaps early, and keep the product moving without waiting on formal handoffs.

## Senior Front-End Engineer

Intuit (via Cydio Group contract) - Remote | *Feb 2023 – Jul 2023*

- Built proof-of-concept UI features for Komposer against REST APIs.
- Focused on state consistency in complex workflows.
- Provided front-end guidance to full-stack engineers.

## Senior Front-End Engineer

SparkCognition (via MAANA acquisition) - Bellevue, WA & Remote | *Feb 2019 – Jan 2023*

- Built UI for a computational knowledge graph platform.
- Designed a Bayesian Network authoring interface.
- Integrated front-end features with GraphQL backends.
- Used MobX for application and data state.
- Refactored UI for consistency and maintainability.
- Introduced libraries for state, diagramming, and testing.
- Mentored interns on front-end engineering fundamentals.

## SELECTED EARLIER EXPERIENCE

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- **Microsoft:** UI and platform engineering for developer tools
- **eBay:** Front-end architecture for global shipping workflows
- **Southern Company:** Reusable component strategy and enablement
- **Alaska Airlines:** Customer-facing e-commerce UI
- **CommerceHub:** Backend service design and delivery
- **Magenic Technologies:** Performance, usability, and code quality consulting

## PERSONAL PROJECTS

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- **Clinical Differential Diagnosis Assistant:** Claude-powered tool helping mental health clinicians produce more complete diagnoses --- hybrid LLM extraction plus deterministic rule engine, schema-driven React UI. Built with a clinical expert partner (20+ years).
- **Echo Chamber:** Desktop application for creating and running teams of AI agents via the Anthropic API --- React, TypeScript, Express backend, streaming chat interface.
- **Field Kit:** Offline-first mapping platform for locating USGS survey markers without cell service --- React PWA, IndexedDB, service workers.

## EDUCATION

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**Bachelor of Science**, Computer Science and Music (New College)  
University of Alabama, Tuscaloosa, AL

## CERTIFICATIONS

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- Foundations in Design Thinking - IDEO U
- AI Design Academy - UX Writing Hub
- Fundamentals of UX Writing - UX Content Collective